

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0979

Thomas Hill Station 230 kV
April 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0979. The Interconnection Customer proposes a 141.8 MW wind facility interconnecting at the Thomas Hill Station 230 kV substation in the Indiana area. The request was received on April 09, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 410 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,986,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BLUE RAVINE-SCHOOL – HENRY COUNTY 230 KV line	BLU RAVINE-SCHOOL – DSOTO 230 KV	205.0	507	17.7
HENRY COUNTY 500/230kV Transformer	E METHUEN – LONETREE SUBSTATION 500 KV	185.1	894	20.6
MURDOCK SUBSTATION – SABANERA 230 KV	HENRY COUNTY – E METHUEN 500 KV L	153.4	731	11.3
GRAY'S FLAT – EAST METHUEN 230 KV	MADELINE SUBSTATION – EAST METHUEN 230 KV	149.3	808	13.0
DESOTO SUBSTATION – BLUE RAVINE-SCHOOL 230 KV	BLUE RAVINE-SCHOOL – HENRY COUNTY 230 KV line	147.8	643	14.8
MADELINE SUBSTATION – BARAGA VLG 230 KV	GRAYS FLAT – E METHUEN 230 KV L	146.6	514	9.7
EAST METHUEN 500/230kV Transformer Bank	SWAN VLY – TREGO 500 KV	143.1	1,605	23.3
JAMES DE YOUNG – ADIRONDACK 230 KV	ST. LAWRENCE – JAMES DE YOUNG 230 KV	135.2	748	9.6
CARLEY RD – JAMES DE YOUNG 230 KV	E METHUEN 500 kV/230 kV TX	133.9	647	9.8
ADIRONDACK 500-230 kV Autotransformer	EAST METHUEN 500/230kV XFMR	133.1	1,026	12.0
CORRIDOR – LARSON T 230 KV	TREGO – BEAVER SUBSTATION 500 KV L	128.6	640	7.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MURDOCK – HENRY COUNTY 230 KV line	E METHUEN – HENRY COUNTY 500 KV Line	127.8	848	7.7
HEBO – BURRIS PIT 230 KV	TREGO – BEAVER 500 KV	127.5	933	10.3
BURRIS PIT – JAMES DE YOUNG 230 KV	BEAVER – TREGO 500 KV	126.7	955	10.3
HEBO – WINTHROP 230 KV	HEBO – BARRETT 230 KV	124.7	629	8.0
WALDO SUBSTATION 500 kv/230 kv Transformer Bank	TREGO – BEAVER 500 KV	123.2	881	7.4
LONETREE 500-230 kv Auto	LONETREE – EAST METHUEN 500 KV	121.6	811	10.1
MADELINE – E METHUEN 230 KV	E METHUEN – GRAY'S FLAT 230 KV	121.0	1,074	13.8
JOSO – ONANCOCK DIESEL GENERATION FACILITY 230 KV	BEAVER – EUFAULA 500 KV Line	117.9	1,721	11.5
BARAGA VLG – GRAY'S FLAT 230 KV	E METHUEN – MADELINE 230 KV	116.8	1,020	13.0
ADAMS – ONANCOCK DIESEL GENERATION FACILITY 230 KV line	BEAVER – TREGO 500 KV L	116.0	2,100	12.0
ADAMS – PICKFORD 230 KV	BEAVER – TREGO 500 KV	115.7	1,089	6.4
PICKFORD – HERALD-TWIN CITIES SUBSTATION 230 KV	TREGO – BEAVER 500 KV	115.4	1,391	7.3
MASON ST – NEW DEPARTURE 230 KV	BEAVER – TREGO SUBSTATION 500 KV	115.0	1,075	5.6
LOOKINGGLASS – CORRIDOR 230 KV	TREGO – BEAVER 500 KV	111.8	763	7.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLUE RAVINE-SCHOOL – HENRY COUNTY 230 KV line	Rebuild line with higher-capacity conductor	\$41,887,000	7% (\$2,991,000)
HENRY COUNTY 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$22,331,000	3% (\$730,000)
MURDOCK SUBSTATION – SABANERA 230 KV	Rebuild line with higher-capacity conductor	\$35,745,000	3% (\$996,000)
GRAY'S FLAT – EAST METHUEN 230 KV	Rebuild line with higher-capacity conductor	\$41,328,000	1% (\$496,000)
DESOTO SUBSTATION – BLUE RAVINE-SCHOOL 230 KV	Rebuild line with higher-capacity conductor	\$46,718,000	5% (\$2,458,000)
MADELINE SUBSTATION – BARAGA VLG 230 KV	Rebuild line with higher-capacity conductor	\$10,899,000	1% (\$134,000)
EAST METHUEN 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$32,602,000	1% (\$452,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
JAMES DE YOUNG – ADIRONDACK 230 KV	Rebuild line with higher-capacity conductor	\$39,518,000	1% (\$531,000)
CARLEY RD – JAMES DE YOUNG 230 KV	Rebuild line with higher-capacity conductor	\$23,634,000	1% (\$261,000)
ADIRONDACK 500/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$38,198,000	2% (\$805,000)
CORRIDOR – LARSON T 230 KV	Rebuild line with higher-capacity conductor	\$23,893,000	1% (\$239,000)
MURDOCK – HENRY COUNTY 230 KV line	Rebuild line with higher-capacity conductor	\$32,438,000	7% (\$2,257,000)
BURRIS PIT – HEBO 230 KV Line	Rebuild line with higher-capacity conductor	\$25,768,000	1% (\$359,000)
BURRIS PIT – JAMES DE YOUNG 230 KV	Rebuild line with higher-capacity conductor	\$43,562,000	1% (\$606,000)
HEBO – WINTHROP 230 KV	Rebuild line with higher-capacity conductor	\$14,775,000	2% (\$217,000)
WALDO SUBSTATION 500 kV/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,232,000	1% (\$299,000)
LONETREE 500-230 kV Auto	Replace transformer with higher-rated unit	\$29,339,000	4% (\$1,325,000)
MADELINE – E METHUEN 230 KV	Rebuild line with higher-capacity conductor	\$31,475,000	1% (\$373,000)
JOSO – ONANCOCK DIESEL GENERATION FACILITY 230 KV	Rebuild line with higher-capacity conductor	\$28,061,000	1% (\$295,000)
BARAGA VLG – GRAY'S FLAT 230 KV	Rebuild line with higher-capacity conductor	\$16,258,000	1% (\$195,000)
ONANCOCK DIESEL GENERATION FACILITY SUBSTATION – ADAMS 230 KV	Rebuild line with higher-capacity conductor	\$54,454,000	1% (\$549,000)
ADAMS – PICKFORD 230 KV	Rebuild line with higher-capacity conductor	\$17,138,000	1% (\$180,000)
PICKFORD – HERALD-TWIN CITIES SUBSTATION 230 KV	Rebuild line with higher-capacity conductor	\$49,218,000	1% (\$501,000)
NEW DEPARTURE – MASON ST 230 KV	Rebuild line with higher-capacity conductor	\$51,907,000	1% (\$508,000)
LOOKINGGLASS – CORRIDOR 230 KV	Rebuild line with higher-capacity conductor	\$22,921,000	1% (\$229,000)
	Total allocated to project		\$17,986,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.